

Towfiqul Islam

Software Engineer | Python – Backend, Automation & AI Tooling | Dhaka, Bangladesh

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Portfolio: <https://towfiq-ul.github.io>

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Professional Summary

Senior Software Engineer with 7+ years of professional experience, primarily in Java/Spring Boot, who applies Python for backend services, automation tooling, and AI/LLM engineering in self-directed and open-source work. Built a full RAG pipeline over S3-hosted documents, a cross-platform encrypted backup daemon, and automation tooling for Claude Code. Brings a strong foundation in distributed systems, clean architecture, and production engineering discipline from a Java/Spring Boot career to Python projects.

Python Highlights

- Built and published a full RAG pipeline (chunking → embeddings → OpenSearch vector search → LLM-grounded answers) over documents stored in AWS S3, documented in a public GitHub repo and a Medium engineering deep-dive
- Built Google Drive Backup Utility, a cross-platform Python backup daemon covering files and databases (MySQL/MariaDB/PostgreSQL) with multiple storage destinations (Google Drive, Dropbox, OneDrive, email, scp), client-side AES-256-GCM encryption before upload, and retry-with-backoff on failed uploads — shipped as a freemium product with a Gumroad-gated Pro tier
- Authored Claude Code automation tooling in Python/Bash: a Stop-hook-based session guardian that detects usage-limit banners and resumes sessions automatically, and a session-transcript exporter that parses Claude Code's JSONL transcripts directly
- HackerRank-certified in Python (Basic); taught Python fundamentals, algorithms, and data structures as a competitive-programming coach

Core Expertise

- **Backend & Scripting:** Python, Bash, REST/API integration, JSON/YAML tooling
- **AI / LLM Engineering:** RAG pipelines, embeddings, vector search (OpenSearch), prompt engineering, LLM integration
- **Cloud & Storage:** AWS S3, Google Drive API, Cloudflare Workers
- **Security:** AES-256-GCM client-side encryption
- **Cross-stack:** Java, Spring Boot, Microservices (primary professional stack); PHP/Laravel

Professional Experience

Note: the roles below were delivered primarily in Java/Spring Boot; Python work is demonstrated through the self-directed and open-source projects listed separately.

Senior Software Engineer – Exabyting

May 2024 – Current

Augmented resource at bKash Limited. Designing microservices, scalable REST APIs, Redis caching, and CI/CD pipelines for a core payment platform handling millions of daily transactions.

Software Engineer – SELISE

November 2021 – April 2024

Extended ThingsBoard into a multi-tenant IoT middleware platform for a Swiss enterprise customer, modeling and optimizing time-series telemetry at a scale of millions of data points.

Software Engineer – Exabyting

December 2019 – November 2021

Developed government and private web applications and services. Worked as partner team with bKash Limited's Solution Engineering Team.

Key Python Projects

- **AI Query System from S3 Data** – RAG pipeline (chunking, embeddings, OpenSearch vector search, LLM-grounded answers); public GitHub repo + Medium deep-dive
- **Google Drive Backup Utility** – Cross-platform encrypted backup daemon, AES-256-GCM, multiple storage destinations, freemium Gumroad product
- **claude-auto-resume** – Python/Bash Claude Code automation; Stop-hook-based session guardian that auto-resumes paused sessions
- **claude-export-session** – Python/Bash Claude Code session-transcript exporter, parses JSONL transcripts directly

Education

B.Sc. in Computer Science & Engineering – International Islamic University Chittagong (2014 - 2019)

Certifications & Awards

- Python (Basic) – HackerRank
- IIUC IUPC 2017 – 2nd Position (Coach)
- EDU Programming Contest 2018 – 5th Position (Coach)
- CSE Week Programming Contest 2016 – 5th Position
- Solved 500+ competitive programming problems